

Product Optimization and Revenue Contribution Analysis for Specialty Coffee Retail: An Interactive Business Intelligence Dashboard Approach

P. Venkatesh

Roll No: AP24311010150

SRM University

Domain: Data Analytics / Business Intelligence

Abstract

The specialty coffee retail industry demands data-driven decision-making to manage diverse product portfolios across multiple store locations. This paper presents the design and implementation of an interactive Sales Intelligence Dashboard for Afficionado Coffee Roasters — a three-location NYC café chain — using Python, Streamlit, and Plotly. The system processes 149,116 transaction records across 80 unique products, applies ABC classification and product segmentation frameworks (Hero, Premium, Long Tail, Underperforming), and exposes actionable insights through real-time filterable visualizations. The dashboard answers three core business questions: which products drive the majority of revenue (Pareto/80-20 analysis), how products are positioned on popularity versus revenue contribution, and where operational focus should be directed across store locations and time periods. Results show that 43 of 80 products (53.8%) are required to reach 80% of total revenue, that the top 20% of products by revenue contribute 37.1% of revenue, and that product popularity and revenue rank are strongly correlated (Spearman $\rho = 0.826$) — a relationship so strong that only four of eighty products fall into the "disagreement" segments (Premium or Long Tail), with the remainder split almost evenly between clear Hero and Underperforming status. The system demonstrates that lightweight open-source tooling can deliver enterprise-grade product intelligence at minimal infrastructure cost.

Keywords: *Business Intelligence, Product Segmentation, ABC Classification, Pareto Analysis, Streamlit, Coffee Retail Analytics, Revenue Optimization*

1. Introduction

The global specialty coffee market has grown substantially over the past decade, with multi-location café operators facing increasing pressure to optimize their product portfolios while managing operational costs. Unlike large retail chains with dedicated BI teams, small-to-medium café operations often lack accessible tools to translate raw transaction data into actionable inventory and pricing decisions.

Product portfolio management in food and beverage retail is inherently complex. A typical specialty café may carry sixty to ninety SKUs spanning beverages of multiple preparation types, loose-leaf teas, packaged chocolates, baked goods, flavouring syrups, and branded merchandise. Without systematic analysis, operators risk over-stocking underperforming items, missing upsell opportunities on high-margin products, and allocating marketing effort inequitably.

This work addresses that gap by building a fully interactive Sales Intelligence Dashboard for Afficionado Coffee Roasters, a three-store NYC operation with locations in Lower Manhattan, Hell's Kitchen, and Astoria. The project contributes: (1) a reproducible analytics pipeline that ingests raw transaction logs and computes KPIs, ABC classes, and product segments live; (2) a web-deployable, zero-installation dashboard that non-technical stakeholders can use without SQL or spreadsheet skills; and (3) an applied demonstration of Pareto analysis, ABC classification, and quadrant-based product segmentation within a

real-world retail context, using a genuine 80-product, 149,116-transaction dataset rather than a toy example.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset. Section 4 presents the system design and methodology. Section 5 discusses results and business insights. Section 6 addresses limitations, and Section 7 concludes with future directions.

2. Related Work

Business Intelligence in Retail

Modern BI platforms such as Tableau and Power BI provide drag-and-drop analytics, but they require paid licenses and proprietary data connectors. Streamlit has emerged as a strong open-source alternative for data scientists who want to convert Python analysis scripts directly into shareable, interactive web applications without front-end development [7].

ABC Classification

The ABC inventory classification method, rooted in the Pareto principle, segments items into three classes based on cumulative contribution to a key metric — typically revenue or usage value. Class A items receive the highest management attention; Class B items are moderate priority; Class C items are lowest priority [8] [9]. The method has been widely validated in manufacturing, retail, and healthcare supply chains, and its threshold boundaries (commonly 70/20/10 or 80/15/5 splits of cumulative revenue) are adapted to the specific business context rather than fixed by convention.

Product Portfolio Segmentation

Beyond ABC classification, two-dimensional product segmentation using popularity (units sold) versus revenue has been used in retail to identify distinct strategic groups [2]. Products with high revenue and high popularity are “Heroes” — the core of the business. High-revenue, lower-popularity products are “Premium” — niche but valuable. High-popularity, lower-revenue products are “Long Tail” — high-volume but margin-thin. Low on both dimensions are “Underperforming” — candidates for discontinuation or repositioning.

Pareto Analysis in Food and Beverage

The 80/20 rule has been applied extensively in food-and-beverage contexts to identify the subset of menu items generating the majority of revenue [1][3]. Menu engineering, a related discipline, combines contribution margin with menu-mix percentage to classify items as Stars, Plowhorses, Puzzles, and Dogs — a framework this project's segmentation model partially parallels, substituting revenue share for margin data that was not available in the source dataset.

3. Dataset Description

The dataset comprises three related tables derived from a one-year transaction log for Afficionado Coffee Roasters.

3.1 Transaction-Level Dataset (*coffee_features.csv*)

Rows: 149,116 individual transaction line items, each representing one product sold in one transaction. Key fields: `transaction_id`, `store_location`, `product_id`, `product_detail`, `product_category`, `product_type`, `transaction_qty`, `unit_price`, `revenue`, `hour`, and `time_slot`.

3.2 Product Summary (*product_summary.csv*)

Rows: 80 unique SKUs, each with `total_revenue`, `total_qty`, `transaction_count`, `avg_unit_price`, `revenue_pct`, `abc_class`, `revenue_rank`, `volume_rank`, and segment already computed. Categories: Coffee, Tea, Coffee Beans, Loose Tea, Bakery, Drinking Chocolate, Packaged Chocolate, Flavours, and Branded — nine categories in total.

3.3 Store Summary (*store_summary.csv*)

Rows: 3 records, one per store, with `total_revenue`, `total_qty`, `transaction_count`, `avg_order_value`, and `revenue_share_pct`. Total system revenue across all three stores is \$698,812.

Table 1: Dataset Summary Statistics

Metric	Value
Total Revenue	\$698,812
Total Transactions	149,116
Total Units Sold	214,470
Unique SKUs	80
Store Locations	3
Product Categories	9
Average Order Value	\$4.69
Date Range	1 year

4. System Design and Methodology

4.1 Architecture

The application follows a single-tier, stateless architecture consisting of three layers. The Data Layer loads the three CSV files into Pandas DataFrames at startup and caches them with `@st.cache_data` to avoid redundant I/O on filter interactions. The Computation Layer recomputes all metrics — KPIs, rankings, cumulative revenue percentages, and segment-level aggregates — live from the filtered transaction subset whenever the user changes a sidebar filter. The Presentation Layer renders interactive Plotly charts (bar, donut, scatter, and Pareto combination charts) and a configurable, searchable data table through Streamlit widgets.

[CSV Files] → [Pandas DataFrames] → [Aggregation & Segmentation] → [Plotly Charts + Streamlit UI]

4.2 Data Processing Pipeline

On application load, the transaction fact table is joined to the product master (for the pre-computed ABC class and segment labels) on `product_id`. Store name normalization handles a capitalization inconsistency present in the source export (“Hell’S Kitchen” vs. “Hell’s Kitchen”).

After sidebar filters are applied, a product-level aggregation is computed by grouping the filtered transactions on `product_id`, `product_detail`, `product_category`, and `product_type`, summing revenue and units and counting transactions. Derived fields added to this aggregation include `avg_price` (revenue ÷ units), `revenue_pct` (product revenue ÷ total filtered revenue), `cum_revenue_pct` (cumulative revenue_pct sorted by revenue descending), `revenue_rank` and `volume_rank` (via `Pandas rank()`), and `rank_delta` (`volume_rank - revenue_rank`), which reveals products that are popular but under-priced relative to demand, or vice versa.

4.3 ABC Classification

ABC classification was pre-computed on the full dataset (stored in `product_summary.csv`) using cumulative revenue thresholds of 70% and 90% rather than the more commonly cited 80%/95% split: Class A products are those within the top cumulative 70% of revenue when products are ranked from highest to lowest; Class B spans the next band from 70% to 90%; Class C is the remaining 90% to 100% of cumulative revenue (the long tail). In the full dataset, 35 products carry Class A designation, 17 carry Class B, and 28 carry Class C.

4.4 Product Segmentation (Quadrant Model)

Each product is assigned to one of four segments based on its position relative to the median units sold and median revenue across all products in the current filter selection:

Table 2: Product Segmentation Quadrant Model

Segment	Revenue vs. Median	Units vs. Median	Strategy
Hero	Above	Above	Protect and promote
Premium	Above	Below	Upsell, maintain margin
Long Tail	Below	Above	Review pricing
Underperforming	Below	Below	Rationalize or discontinue

4.5 Pareto Analysis

The Pareto chart renders products sorted by descending revenue on the primary y-axis (bar chart) with a cumulative revenue percentage line on the secondary y-axis. A dashed reference line at 80% highlights the Pareto threshold — the minimum number of products needed to account for 80% of revenue.

4.6 Dashboard Interface

The dashboard is structured into four tabs. Overview presents KPI cards, revenue by category (horizontal bar), revenue by store (donut chart), revenue by time slot (bar), and a peak-hour caption. Product Analysis presents Top-N by revenue, Top-N by units, Bottom-N by revenue, and a popularity-vs-revenue scatter plot with quadrant reference lines. Pareto & Segmentation presents the Pareto concentration chart, an ABC classification bar chart, and a segment revenue bar chart. Data Table presents a searchable, downloadable drill-down table with all computed metrics. Global sidebar filters — store, category, product type, and a Top-N slider — affect all four tabs simultaneously.

5. Results and Business Insights

5.1 Revenue Concentration (Pareto Finding)

Analysis of the 80-product portfolio shows a real but moderate Pareto concentration: **43 of 80 products (53.8% of the portfolio)** are required to reach 80% of total revenue, and the top 20% of products (16 items) contribute **37.1% of revenue**. This is a meaningfully less extreme concentration than the textbook 80/20 split — the portfolio does not have a small handful of products carrying the business, but it does have a clearly identifiable core: the top ten products alone (12.5% of the menu) already account for 25.4% of revenue.

Table 3: Top 10 Revenue-Generating Products

Rank	Product	Revenue (\$)	Cumulative %
1	Sustainably Grown Organic Lg	\$21,152	3.0%
2	Dark Chocolate Lg	\$21,006	6.0%
3	Latte Rg	\$19,112	8.8%
4	Cappuccino Lg	\$17,642	11.3%
5	Morning Sunrise Chai Lg	\$17,384	13.8%
6	Latte	\$17,258	16.3%
7	Jamaican Coffee River Lg	\$16,481	18.6%
8	Sustainably Grown Organic Rg	\$16,234	20.9%
9	Cappuccino	\$15,998	23.2%
10	Brazilian Lg	\$15,110	25.4%

Notably, the two highest-revenue products are large-size drinking chocolates, not coffees — an insight that would be non-obvious without systematic analysis, and one that a manager relying on intuition about a “coffee shop” would be unlikely to surface unprompted.

5.2 ABC Classification Results

Class A (35 products, 69.5% of revenue): dominated by Coffee (15 of 21 products) and Tea (14 of 16 products), plus all four Drinking Chocolate items — brewed beverages across regular and large sizes. These products demand priority shelf space, consistent quality control, and promotional investment.

Class B (17 products, 20.3% of revenue): a mix of smaller-size brewed beverages and bakery items — solid contributors warranting routine monitoring rather than intervention.

Class C (28 products, 10.2% of revenue): Loose Tea (8 of 8 products), Coffee Beans (9 of 10), Flavours (4 of 4), Packaged Chocolate (3 of 3), and Branded merchandise (3 of 3). These have low individual revenue contribution but may serve brand-image, gifting, or cross-sell functions that pure revenue analysis does not capture.

5.3 Product Segmentation Insights

Segmenting the full portfolio by units sold and revenue against their respective medians produces a strikingly bimodal split: **38 Hero products and 38 Underperforming products, against only 2 Premium and 2 Long Tail products combined**. The entire Premium segment consists of just Civet Cat and Chocolate Croissant; the entire Long Tail segment consists of just Our Old Time Diner Blend Sm and Columbian

Medium Roast Sm. This asymmetry is directly explained by a Spearman rank correlation between revenue_rank and volume_rank of $\rho = 0.826$ ($p < 0.001$) — because popularity and revenue move together so closely across this menu, very few products end up "disagreeing" (popular-but-cheap, or rare-but-expensive); most products are simply either broadly successful on both dimensions or broadly weak on both.

This is a materially different — and more actionable — finding than a vague quadrant description would suggest: it means portfolio strategy for this business is less about hunting for hidden gems in a Premium or Long Tail segment (there are almost none) and more about a binary decision for most of the menu: protect the Heroes, and seriously reconsider the Underperformers.

5.4 Store Performance

Revenue distribution across the three stores is remarkably uniform:

Table 4: Store-Level Performance

Store	Revenue	Share	Avg Order Value
Hell's Kitchen	\$236,511	33.8%	\$4.66
Astoria	\$232,244	33.2%	\$4.59
Lower Manhattan	\$230,057	32.9%	\$4.81

The near-equal distribution suggests consistent customer demand across NYC neighbourhoods and indicates that product-level insights derived from aggregate data apply broadly across all three locations rather than being an artefact of one dominant store. Average order values are similarly close, with Lower Manhattan showing a modest premium (\$4.81 vs. \$4.59–\$4.66 at the other two stores), plausibly reflecting higher-price-point purchases from office-area customers.

5.5 Temporal Patterns

Time-slot analysis shows Morning as the dominant revenue period by a wide margin, contributing 55.6% of total revenue — more than Afternoon (29.3%) and Evening (15.1%) combined. Revenue peaks at 10:00, consistent with the core café use case of commuter and early-office coffee consumption. This pattern gives store managers a concrete basis for staffing schedules and batch-prep planning around the highest-throughput window of the day.

5.6 Rank Delta Analysis

The rank_delta field (volume_rank – revenue_rank) exposes pricing inefficiency. Products where volume_rank is substantially higher than revenue_rank (sold in large quantities but ranking low in revenue) are potential candidates for a price increase relative to demonstrated demand. Conversely, products with revenue_rank far ahead of volume_rank generate strong revenue from comparatively low volume — high-margin niche items whose pricing is likely already well calibrated. Because the two rankings are as tightly correlated as shown in Section 5.3, large rank_delta values in this dataset are the exception rather than the rule, which makes the exceptions that do appear on the dashboard's Product Performance table easy to spot and prioritize.

6. Limitations

- Static classification: the dashboard reads pre-computed ABC classes and segments from the product summary file. As filter selections change, revenue shares recompute dynamically, but the

ABC/segment labels themselves remain the global (unfiltered) classification. A fully live re-classification within each filtered view would occasionally produce different labels than the ones shown.

- No time-series dimension: the dataset covers one aggregated year without month- or week-level granularity, so seasonality and growth trends cannot currently be analyzed.
- Revenue-only, not margin-based: cost-of-goods data is not present in the source system, so all classification and ranking in this paper is revenue-based rather than profit-based; a low-revenue product with very high margin would still be flagged Underperforming here.
- Segmentation threshold sensitivity: the quadrant model uses median values as thresholds, which are themselves sensitive to the current filter selection, meaning a product's segment can shift when filters narrow the dataset. Fixed percentile thresholds defined once on the full dataset would produce more stable labels across filter states.
- No predictive component: the dashboard is descriptive and diagnostic only. Prescriptive recommendations such as reorder quantities or optimal price points would require an additional modelling layer, outside the current scope.

7. Conclusion and Future Work

This paper presented a fully functional, interactive Business Intelligence dashboard for specialty coffee retail product optimization, built with Python, Streamlit, and Plotly. Processing 149,116 transactions across 80 SKUs and 3 store locations, the system implements ABC classification, four-quadrant product segmentation, Pareto concentration analysis, and temporal revenue breakdowns — all accessible through a browser-based interface requiring no installation for end users.

Key findings confirm a real, if moderate, concentration of revenue in this retail context: 43 of 80 products drive 80% of revenue, and the strong correlation between popularity and profitability ($\rho = 0.826$) means the portfolio is dominated by two clear groups — Hero and Underperforming products — with almost no products in the ambiguous Premium or Long Tail middle ground. Brewed beverages and drinking chocolate constitute the Hero segment and Class A backbone, while whole-bean coffees, loose teas, flavour syrups, and branded merchandise populate Class C and the Underperforming segment.

Future work will extend this system in three directions. Live POS integration would replace static CSVs with real-time streams from point-of-sale APIs, enabling continuous KPI refresh. A forecasting module using time-series methods would project weekly revenue by product and store to support proactive inventory planning. A menu engineering overlay would combine contribution-margin data with the existing popularity-revenue framework to produce a full margin-aware quadrant classification (Stars, Plowhorses, Puzzles, Dogs), once cost data becomes available.

The complete source code and dataset are packaged as a deployable Streamlit application, demonstrating that open-source tools can deliver enterprise-quality retail analytics at negligible infrastructure cost — a meaningful capability for small and medium food-and-beverage operators.

References

- [1] M. E. Bayou and L. B. Bennett, "Profitability analysis for table-service restaurants," *Cornell Hotel and Restaurant Administration Quarterly*, vol. 33, no. 2, pp. 49–55, 1992.
- [2] P. Kotler and K. L. Keller, *Marketing Management*, 15th ed. Boston, MA, USA: Pearson Education, 2016.

- [3] V. Pareto, Cours d'Économie Politique. Lausanne, Switzerland: F. Rouge, 1896.
- [4] W. McKinney, Python for Data Analysis, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2017.
- [5] J. D. Hunter, "Matplotlib: A 2D graphics environment," Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.
- [6] M. L. Waskom, "seaborn: statistical data visualization," Journal of Open Source Software, vol. 6, no. 60, p. 3021, 2021.
- [7] Streamlit Inc., "Streamlit Documentation," docs.streamlit.io, 2024. [Online]. Available: <https://docs.streamlit.io>
- [8] J. M. Juran and A. B. Godfrey, Juran's Quality Handbook, 5th ed. New York, NY, USA: McGraw-Hill, 1999.
- [9] H. F. Dickie, "ABC inventory analysis shoots for dollars, not pennies," Factory Management and Maintenance, vol. 109, no. 7, pp. 92–94, 1951.
- [10] Plotly Technologies Inc., "Plotly Python Open Source Graphing Library," plotly.com/python, 2024. [Online]. Available: <https://plotly.com/python/>

Word count: approximately 3,050 words (body) | Submitted in partial fulfilment of coursework requirements — SRM University